DEVICE USER GUIDE

THE OFF-GRID VEHICLE TRACKER, SCREEN BY SCREEN

Every picture in this guide is a real photograph of the actual devices. It walks you from switching a radio on, through every screen and menu you will meet, to switching it off again, with the display covered along the way. If a screen is not in this guide, you do not need it.



GROUP 9 · CAPSTONE PROJECT LAB · TEXAS TECH UNIVERSITY · SPONSOR: DR. RICHARD GALE · 2026-10-05

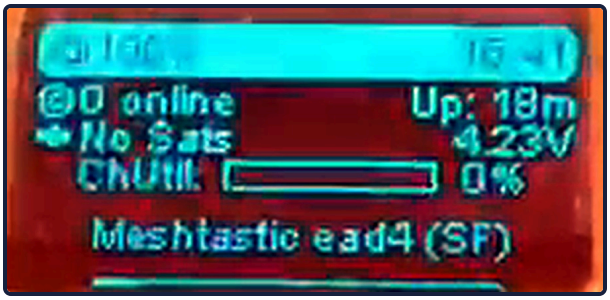
1. Meet the Devices and Their Buttons

Each vehicle carries two devices. The **radio** is the small green box with the antenna; it does all the talking over the air. The **display** is the larger gray dashboard screen; it shows you where your partner is. This guide covers the radio first, because everything starts and ends there, then the display.

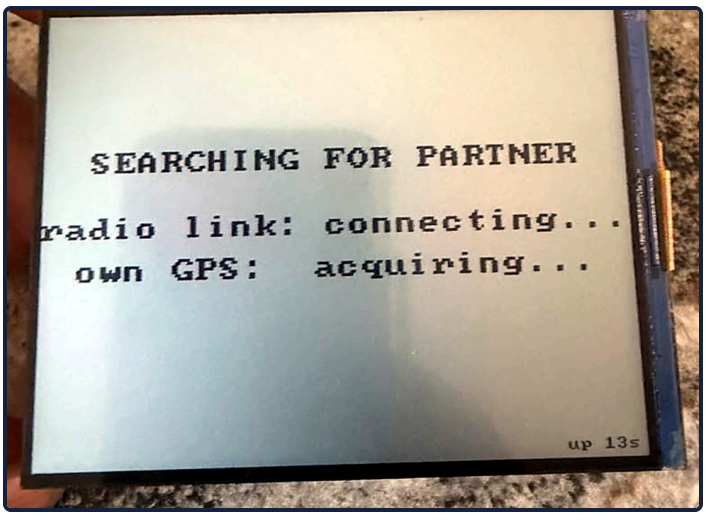
Device	Buttons	What they do
Radio	Two buttons on the top edge	The **user button** (left): a short press moves to the next screen or menu line; a long press opens the action menu for the screen you are on, or confirms the highlighted line in a menu. The **power button** (right): hold it to switch the radio on.
Display	One main button, plus the small BOOT button	A tap moves to the next page. Holding the button resets the tracking data, or clears the message history when the Messages page is showing. The BOOT button asks your partner's radio for a fresh position right now.

> **The one rule that prevents almost every problem:** keep both devices charged, and never run the radio's battery completely flat. A fully drained battery can erase the radio's settings, and then the two radios stop understanding each other until a laptop fixes it.

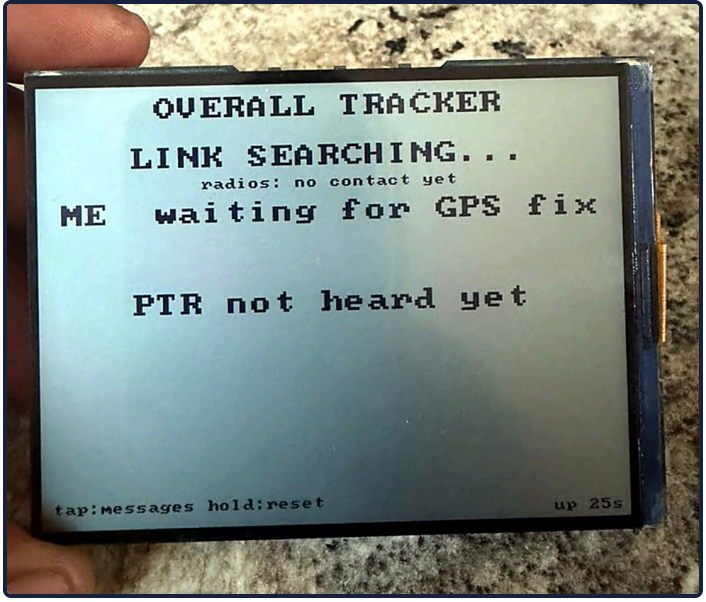
2. Powering On



Hold the radio's power button until the screen lights, then let the radio settle for a minute. This home screen is what normal looks like: battery at the top left, uptime at the top right, and the radio's own name at the bottom, here "Meshtastic ead4 (SF)". Indoors it will say "0 online" and "No Sats"; that is not a fault. Satellites need a view of the sky, and your partner's radio may simply be off.



Power the display at the same time, like turning an ignition key. For its first few seconds it shows this searching screen while it connects to its own radio over Bluetooth and the radio hunts for satellites. The radio only accepts the display's connection during its first minute after power on, which is why the two should always be switched on together. If you miss the window, switch both off and on again together.



Once the display has its radio, the status page fills in piece by piece: "waiting for GPS fix" clears when the satellites arrive, and "PTR not heard yet" clears the first time your partner's radio is heard on the channel. Outdoors with both vehicles on, the whole page is normally live within a minute or two.

3. The Two-Second Pre-Flight Check



Before every trip, short-press the radio's user button until you reach this **LoRa screen**, and read one number: **Freq: 912.875 MHz**. Check it on both radios. If both say 912.875, the radios can hear each other and everything else will follow. If either radio shows a different number (906.875 is the usual wrong answer), that radio has lost its settings and needs the recovery procedure in the Technical Manual before the trip.

While you are here, the rest of the screen should read Role: Client Mute and US/LongFast, exactly as photographed.

4. Reading the Radio's Screens

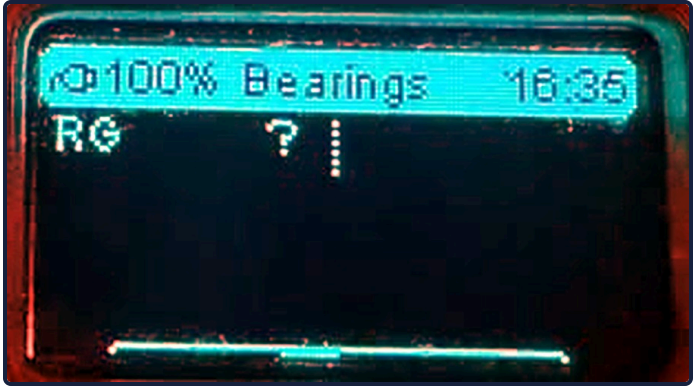
Short presses of the user button walk a loop of status screens. Here is each one and what it tells you.



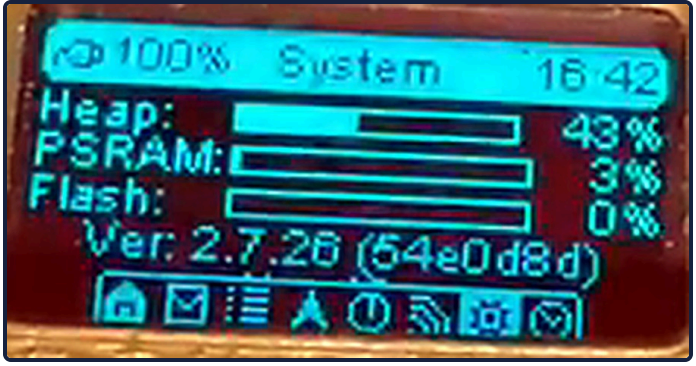
Position. Your own GPS state. "No Sats / No GPS Lock / No Fix" simply means indoors; it fills with coordinates under open sky.



Last Heard. Every radio heard recently and how long ago; here RG was heard 7 minutes ago. This is the radio's own honesty meter.



Bearings. Direction to other radios once positions are known; a question mark means no position for them yet.



System. Memory, firmware version (2.7.26) and uptime. You never need this page except when someone asks what version you run.



Clock. Time and date arrive from the GPS satellites; nothing to set.



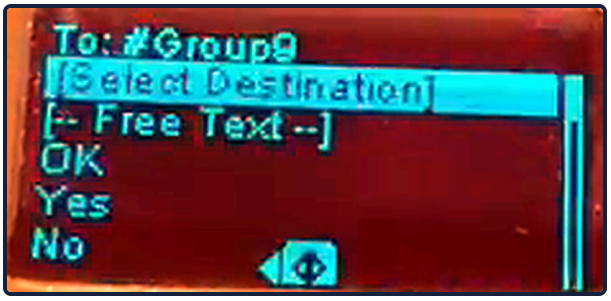
Channel chat. The running conversation on the Group9 channel, newest at the bottom.

5. Sending a Message from the Radio

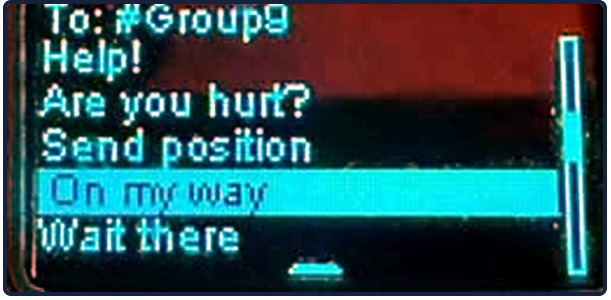
The radio sends ready-made messages so nobody types while driving. The whole flow is five screens, and every press is the same user button: short press to move, long press to choose.



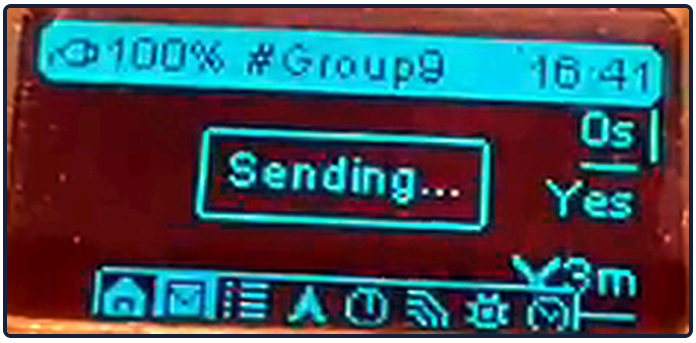
Step 1. From the chat screen, a long press opens the **Message Action** menu. Short-press down to Reply (or View Chats to browse), then long-press to choose.



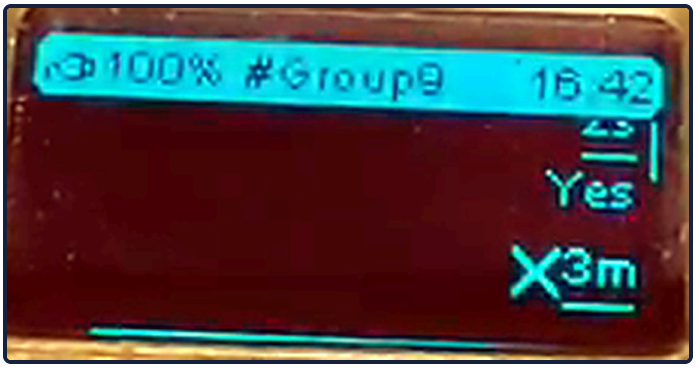
Step 2. Choose the destination. **#Group9** is the shared channel both vehicles hear, and it is the one you want. Skip Free Text; it is for typing with a phone attached, which no one does while driving.



Step 3. Scroll the preset list with short presses. The 21 messages cover the whole conversation: Help!, Are you hurt?, Send position, On my way, Wait there, Turn left, Turn right, Stop, and the rest. Long-press the one you want.



Step 4. The radio shows "Sending..." for a moment as the message goes out over the air.



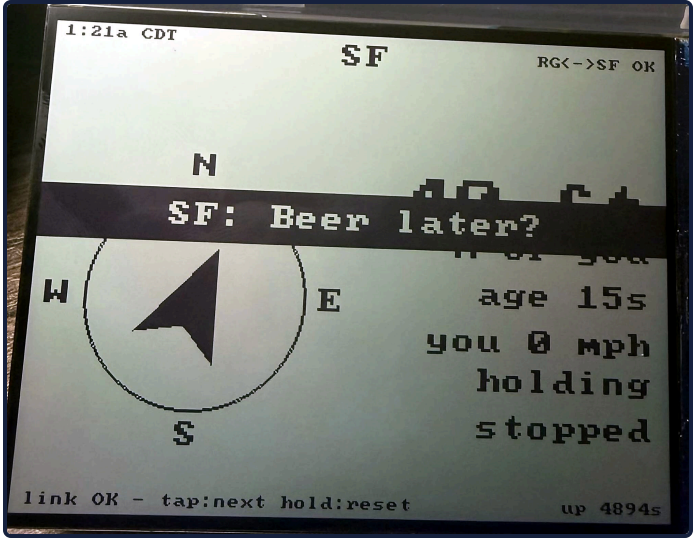
Step 5. The message joins the thread with a delivery mark, and it pops up on your partner's display within a few seconds. An X next to a message means no radio has confirmed it yet.



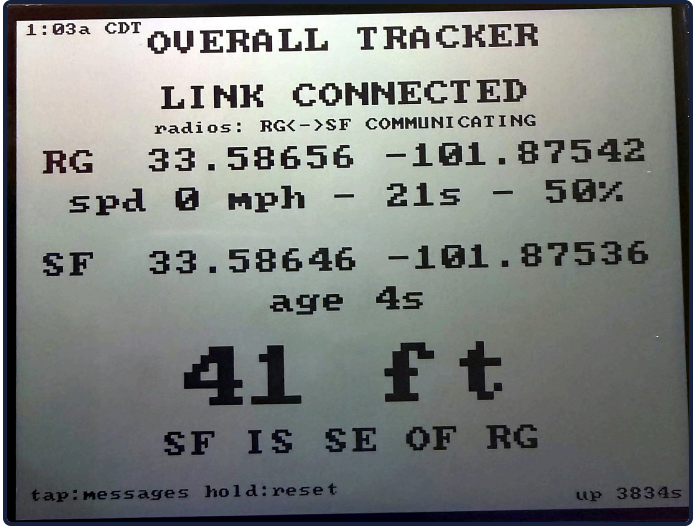
Replying is even shorter. When a message arrives, long-press on the chat screen and choose **Reply to Last Msg, With Preset**. The same preset list opens, aimed back at the sender. Every question in the list has its answer in the list: "Are you hurt?" pairs with "I am good" or "Help!", and "Beer later?" pairs with "Yes".

6. The Display While You Drive

You never need to touch the display while driving; it updates itself twice a second. A tap cycles its three pages.



The compass page. The arrow points at your partner; while driving, straight up means dead ahead. The big number is the distance. An arriving message appears as a dark banner, here "SF: Beer later?", for eight seconds.



The status page. Both vehicles' coordinates and data ages, the partner radio's battery, the distance in big digits, and the direction in words: "SF IS SE OF RG".



The Messages page keeps the last few received texts, newest first. "no messages" is its honest empty state. Holding the button on this page clears the history; holding it on any other page resets the tracking data, which is harmless because fresh data rebuilds in seconds.

Want your partner's position right now? Press the display's BOOT button. Their radio answers within seconds, and the age number on your screen drops to nearly zero.

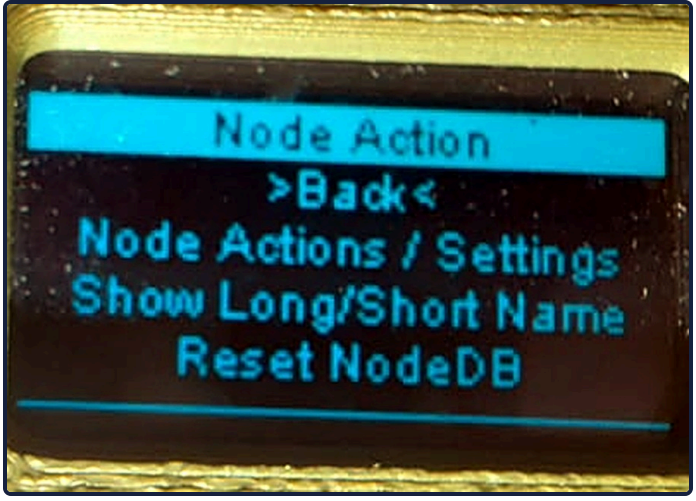
7. Menus to Use, Menus to Leave Alone



Safe and useful: Home Action. A long press on the home screen offers **Send Position**, which broadcasts your position immediately, and Sleep Screen, which dims the screen to save battery. Both are harmless.



Safe: GPS Action and Clock Action adjust formats only. Leave Smart Position switched on; it is what saves airtime.



Troubleshooting only: Node Action. Reset NodeDB wipes the list of known radios. Use it only when the Technical Manual says so.



Do not change anything under LoRa Actions. Device Role, Radio Preset, Frequency Slot and LoRa Region are the exact settings that let the two radios hear each other. Changing any of them on one radio makes the pair go silent with no error message anywhere. The same caution applies to Bluetooth Toggle under System Action: switching Bluetooth off strands your display.

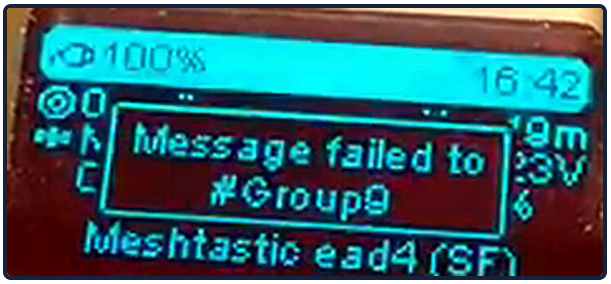
8. Powering Off and Charging



Long-press on the home screen, open System Action, and choose **Reboot/Shutdown**. The radio says "Shutting Down..." and switches off cleanly. Prefer this over letting the battery die: a clean shutdown always preserves the settings.

Charge both devices over USB-C after every trip. The radio holds its charge for months when switched off. The display has no battery of its own worth noting; power it from the vehicle or a USB source.

9. When Something Looks Wrong



"Message failed" means nobody acknowledged the message, usually because the partner's radio is off, out of range, or still booting. The message is not lost forever; send it again once the Last Heard screen shows the partner recently.

What you see	What it means	What to do
0 online, No Sats indoors	Normal. No sky view, partner off.	Nothing. Take it outside and switch the partner on.
Frequency is not 912.875	The radio lost its settings, usually after a dead battery.	Recovery procedure in the Technical Manual (five minutes with a laptop).
Display stuck on "SEARCHING FOR PARTNER"	It missed the radio's connection window.	Switch radio and display off, then on together.
Partner on screen but data age climbing	Distance or terrain is blocking the link.	It heals itself as you get closer. The screen never pretends.
Anything else odd	Electronics being electronics.	Power-cycle both devices: off, count to five, on together.

10. Quick Reference

I want to...	Do this
Start up	Charge check, antennas on, power radio and display together, confirm the display connects.
Pre-flight	Short-press to the LoRa screen on both radios: both must read 912.875.
Send a preset message	Long press on chat, Reply or destination #Group9, pick the preset, long press.
Reply fast	Long press, Reply to Last Msg, With Preset.
Broadcast my position now	Long press on home, Send Position.
Get partner's position now	Press BOOT on the display.
Shut down	Long press on home, System Action, Reboot/Shutdown.
Fix almost anything	Power-cycle both devices together; check 912.875.

Off-Grid Vehicle Tracker · Group 9, Texas Tech University ECE Capstone · Companions: User Manual (concepts) and Technical Manual (engineering and recovery procedures). All photographs are of the actual Group 9 hardware, radio firmware Meshtastic 2.7.26.